

AMBUJ TIWARI

Kolkata, West Bengal

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Summary

Software Engineer with 3.9 years of experience in Java development, specializing in Spring Boot, Microservices, REST APIs, Angular, and AWS cloud technologies. Experienced in designing scalable backend systems, developing cloud-native applications, and implementing serverless architectures. Hands-on experience with Generative AI, LLMs, RAG, and LangChain. Skilled in performance optimization, debugging, and delivering robust enterprise solutions in Agile environments.

TECHNICAL SKILLS

Languages: Java (8, 21), Python, C

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Microservices, REST APIs

Frontend: Angular, TypeScript, JavaScript, HTML5, CSS3

Databases: MySQL, DynamoDB, FAISS

Cloud & DevOps: AWS (Lambda, API Gateway, EC2, S3, SQS, Step Functions, Bedrock), Docker, CI/CD

Tools: Git, Postman, IntelliJ IDEA, VS Code, Linux

AI/GenAI: RAG, LLMs, LangChain

WORK EXPERIENCE

Cognizant

Mar 2022 – Present

Multiple Roles

Kolkata / Remote

Associate – Backend Developer

Aug 2024 – Present

- Enhances and maintains **Spring Boot microservices**, reducing production issues by **15%**.
- Designed and implemented APIs using a **Design-First approach** and integrated **SOAP-based (WSDL) services** for seamless communication with enterprise systems.
- Developed optimized service interactions using **CompletableFuture** for parallel processing and **Spring WebFlux WebClient** with **OpenAPI-generated clients**, ensuring reactive communication, type safety, and reduced manual integration effort.
- Builds **AWS serverless workflows** (Lambda, API Gateway, DynamoDB), reducing manual effort by **25%**.
- Performs **performance tuning and debugging** to improve system reliability.
- Collaborates with cross-functional teams to deliver features across sprints.

Programmer Analyst – Full Stack Developer

Mar 2024 – Nov 2024

- Engineered backend services using **Java and Spring Boot** for insurance workflows.
- Integrated Angular modules with backend APIs across **5+ screens**.
- Optimized SQL queries, improving response time by **20%**.
- Participated in Agile ceremonies ensuring smooth delivery.

Programmer Analyst Trainee – Frontend Developer

Aug 2022 – Mar 2024

- Built responsive UI components using Angular and TypeScript.
- Improved UI/UX consistency, reducing defects by **20%**.
- Collaborated with senior developers to deliver features efficiently.

Full Stack Engineering Intern

Mar 2022 – Aug 2022

- Completed training modules using **Java, Spring Boot, Maven, and SQL**.

PROJECT EXPERIENCE

Nexus-P2C-Migration

- Migrated and modernized existing services using **Spring Boot microservices architecture** and standardized API design practices.
- Implemented **reactive communication** using **Spring WebClient** for service-to-service interactions.
- Leveraged **OpenAPI-generated clients** to reduce manual API integration code and improve maintainability.
- Designed concurrent workflows using **CompletableFuture** for optimized service execution and reduced processing time.

- Integrated monitoring and distributed tracing through **Dynatrace** for performance visibility and issue diagnosis.
- **Supported deployment activities through CI/CD pipelines, maintaining quality standards through SonarQube checks and code coverage validation**

Engineering Excellence

- Led migration of microservices to Java 21, improving performance.
- Created lightweight dependencies reducing external reliance.
- Executed testing achieving **90%+ coverage**.

KT-Buddy

- Developed **KT-Buddy**, an AI-powered **Knowledge Transfer Assistant** that reduces onboarding effort by transforming large codebases into interactive and explainable knowledge systems.
- Built repository ingestion workflows with **GitHub URL validation and automated cloning**, enabling rapid analysis of existing projects.
- Integrated **Anthropic Claude AI** to generate **Javadoc summaries, inline code explanations**, and contextual insights explaining both **what the code does and why it exists**.
- Implemented **RAG-powered question answering** to allow developers to ask natural language questions and receive responses grounded in the actual source code.
- Generated **project overview reports, architecture summaries, and flow visualizations** using **ReportLab**, improving project understanding and documentation.
- Leveraged **Tree-sitter** for source code parsing and analysis, enabling structured understanding of large repositories.
- **Designed and deployed the application using Python, Streamlit, and modern AI workflows for an accessible end-to-end user experience.**

GenAI Visionaries

- Built serverless pipeline with **18+ AWS Lambdas**.
- Integrated AWS services reducing effort by **60%**.
- Applied FAISS-based semantic search with LLMs.
- Improved developer productivity via automation.

G-Synergy

- Enhanced Spring Boot microservices for insurance workflows.
- Engineered REST APIs for integrations.
- Optimized database performance by **20%**.

MNX Rationalization

- Built **responsive UI components** using Angular and TypeScript across **5+ screens**, improving user experience and consistency.
- Improved UI/UX and resolved compatibility issues.

EDUCATION

RCC Institute of Information Technology

B.Tech in Electronics and Communication Engineering

2018 – 2022

CGPA: 8.7

CERTIFICATIONS

- Claude Certified Architect Foundations
- Spring Professional Certification
- MySQL and JDBC
- LangChain for Beginners